



RIPHAH
FACULTY OF REHABILITATION
& ALLIED HEALTH SCIENCES

**DATABASE DOCUMENT
(PDF FILE)**

**FOR
ONLINE GROCERY MART PLATFORM**

SUBMITTED BY: Anshrah, Bazla Kaleem

SUBMITTED TO: Sir Fawad

SAP ID: 66731, 66196

SECTION: BSSE-3A

SUBJECT: Database

ONLINE GROCERY MART PLATFORM

INTRODUCTION

The Online Grocery Mart System is a console-based application that we created in C++ for managing Online Grocery Shopping. It allows user to browse products, add to cart, place order and check their order history. It also allows admin to view all inventory, add new product, update product price, change stock level, check customer orders and checkout. This project shows how programming and data structure ideas can be used in real-world applications. The system stores data permanently using different methods like binary search tree for products, queue for orders, hash tables for users, stack for admin logs and file handling is also used in it. It also ensures fast searching, secure login and multiple ways to pay.

OBJECTIVES

The objective of the project is to create a user-friendly application for Online Grocery Management System.

The system aims to:

- Provide convenience (24/7 shopping with home delivery).
- Provide user-friendly interface.
- Secure payments & personalized recommendations.
- Easy ordering & fast checkout.
- Provide real-time inventory.

PROBLEM DESCRIPTION

What?

The Online Grocery Mart is a console-based application that we created in C++ for managing Online Grocery Shopping. Users can browse products, add to cart, place order and check their order history. It also allows admin to view all inventory, add new product, update product price, change stock level, check customer orders and checkout. This project shows how programming and data structure ideas can be used in real-world applications.

Why?

Traditional shopping requires our physical appearance is necessary, but the advantage of Online Shopping is that we can easily shop from anywhere at anytime in the world. It is a user-friendly interface that helps to solve in real-world problems.

How?

The system stores data permanently using different methods:

- Binary search tree for products
- Queues for orders
- Hash table for users
- Stack for admin logs
- File handling is also used in it. It also ensures fast searching, secure login and multiple ways to pay.

USERS

- New Users: Can signup and create account.
- Registered Users: Browse products, add to cart, place order and view their past history.
- Admin: view all inventory, add new product, update product price, change stock status, check all customer orders and checkout.

TABLES

```
CREATE DATABASE OnlineGroceryMart;  
USE OnlineGroceryMart;
```

```
CREATE TABLE Customer (  
    customerID INT PRIMARY KEY AUTO_INCREMENT,  
    name VARCHAR(100),  
    email VARCHAR(100) UNIQUE,  
    password VARCHAR(100),  
    address VARCHAR(255),  
    phoneNumber VARCHAR(20)  
);
```

```
CREATE TABLE Product (  
    productID INT PRIMARY KEY AUTO_INCREMENT,  
    name VARCHAR(100),  
    price FLOAT,  
    category VARCHAR(100),  
    quantity INT,  
    stockstatus VARCHAR(50)  
);
```

```
CREATE TABLE Cart (  
    cartID INT PRIMARY KEY AUTO_INCREMENT,  
    customerID INT UNIQUE,  
    FOREIGN KEY (customerID) REFERENCES Customer(customerID)  
);
```

```
CREATE TABLE CartItem (  
    cartItemID INT PRIMARY KEY AUTO_INCREMENT,  
    cartID INT,  
    productID INT,  
    quantity INT,  
    price FLOAT,  
    FOREIGN KEY (cartID) REFERENCES Cart(cartID),  
    FOREIGN KEY (productID) REFERENCES Product(productID)  
);
```

```
CREATE TABLE Orders (  
    orderID INT PRIMARY KEY AUTO_INCREMENT,  
    customerID INT,  
    orderDate DATE,  
    status VARCHAR(50),  
    totalAmount FLOAT,  
    FOREIGN KEY (customerID) REFERENCES Customer(customerID)  
);
```

```
CREATE TABLE Payment (  
    paymentID INT PRIMARY KEY AUTO_INCREMENT,  
    orderID INT UNIQUE,  
    method VARCHAR(50),  
    status VARCHAR(50),  
    FOREIGN KEY (orderID) REFERENCES Orders(orderID)  
);
```

```
CREATE TABLE DeliveryPerson (  
    deliveryID INT PRIMARY KEY AUTO_INCREMENT,  
    name VARCHAR(100),  
    contact VARCHAR(50)  
);
```

```
CREATE TABLE Admin (  
    adminID INT PRIMARY KEY AUTO_INCREMENT,  
    username VARCHAR(100),  
    password VARCHAR(100)  
);
```

```
CREATE TABLE StaffSupplier (  
    staffID INT PRIMARY KEY AUTO_INCREMENT,  
    name VARCHAR(100),  
    role VARCHAR(50)  
);
```

```
INSERT INTO Customer (name, email, password, address, phoneNumber) VALUES  
( 'Shaaz', 'shaaz547@gmail.com', 'shaaz547', 'Lahore', '03045577257'),  
( 'Aqeel', 'aqeelwasti@gmail.com', 'aqeel547', 'Lahore', '03111234567'),  
( 'Beenish', 'beenishwasti@gmail.com', 'beenish745', 'Karachi', '03056784567'),  
( 'Ayesha', 'ayesha987@gmail.com', 'ayesha987', 'Hyderabad', '03241013675'),  
( 'Zain', 'zain524@gmail.com', 'zain524', 'Faisalabad', '031442672567');
```

```
INSERT INTO Product (name, price, category, quantity, stockstatus) VALUES  
( 'Yougurt', 200, 'Dairy', 50, 'In Stock'),  
( 'Nuggets', 1000, 'Frozen Items', 30, 'In Stock'),  
( 'Bread', 90, 'Bakery', 150, 'In Stock'),  
( 'Eggs', 300, 'Dairy', 120, 'In Stock'),  
( 'Flour', 1200, 'Grains', 40, 'In Stock');
```

```
INSERT INTO Cart (customerID) VALUES  
(1),  
(2),  
(3);
```

```
INSERT INTO CartItem (cartID, productID, quantity, price) VALUES  
(1, 5, 1, 1200),  
(2, 2, 2, 1000),  
(3, 3, 2, 180);
```

```
INSERT INTO Orders (customerID, orderDate, status, totalAmount) VALUES
(1, '2025-01-10', 'Confirmed', 1200),
(1, '2025-05-10', 'Confirmed', 1000),
(2, '2025-08-11', 'Pending', 500);
```

```
INSERT INTO Payment (orderID, method, status) VALUES
(1, 'COD', 'Paid'),
(2, 'Mobile Wallet', 'Paid'),
(3, 'Card', 'Pending');
```

```
INSERT INTO StaffSupplier (name, role) VALUES
('Bilal', 'Customer Support'),
('Hassan', 'Customer Support'),
('Hina', 'Supplier');
```

SCHEMA

```
Customer(
    customerID PK,
    name,
    email,
    password,
    address,
    phoneNumber
)
```

```
Product(
    productID PK,
    name,
    price,
    category,
    quantity,
    stockstatus
)
```

```
Cart(
    cartID PK,
    customerID FK → Customer(customerID)
)
```

```
CartItem(
    cartItemID PK,
    cartID FK → Cart(cartID),
    productID FK → Product(productID),
    quantity,
    price
)
```

```
Orders(
    orderID PK,
    customerID FK → Customer(customerID),
    orderDate,
    status,
    totalAmount
)
```

```

Payment(
    paymentID PK,
    orderID FK → Orders(orderID),
    method,
    status
)

```

```

DeliveryPerson(
    deliveryID PK,
    name,
    contact
)

```

```

Admin(
    adminID PK,
    username,
    password
)

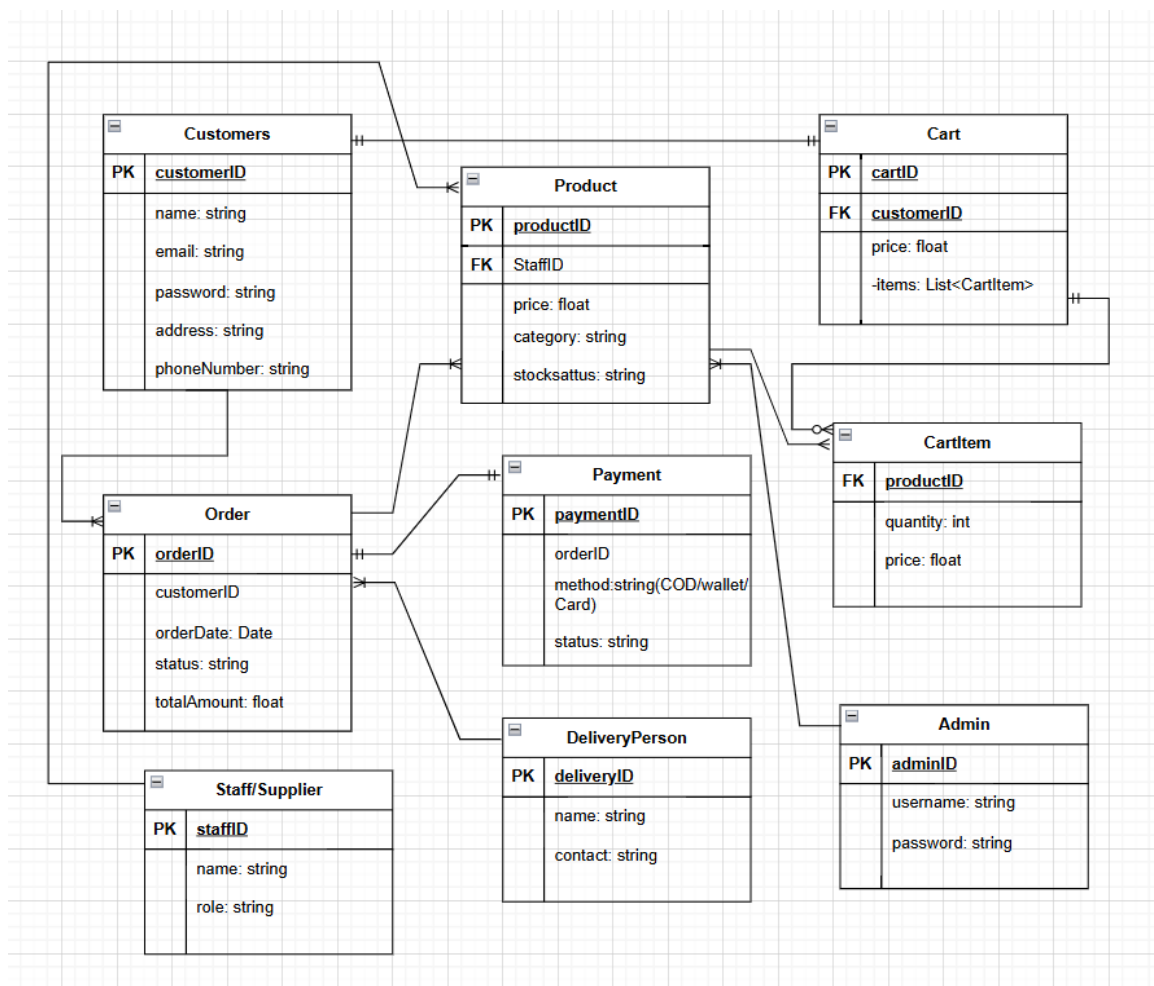
```

```

StaffSupplier(
    staffID PK,
    name,
    role
)

```

ER DIAGRAM



NORMALIZATION

Normalization is used **when designing a database** to organize data properly before implementation. It is used to **remove data redundancy**, **avoid data inconsistency**, and **maintain data integrity**. In this project, the database is normalized up to Third Normal Form (3NF) by separating data into different tables and connecting them using primary and foreign keys such as Customer, Product, Orders, Cart, and Payment

ADVANTAGES

- Provide convenience (24/7 shopping with home delivery).
- Provide user-friendly interface.
- Allow for safe and secure online payments.
- Secure payments
- Personalized recommendations.
- Easy ordering & fast checkout.
- Provide real-time inventory.
- It makes shopping easier and save time.

CONCLUSION

The Online Grocery Mart Platform offers user-friendly, modern and efficient way for people that they can easily shop from anywhere at any time in the world. It also allows admin to view all inventory, add new product, update product price, change stock level, check customer orders and checkout. Overall, this makes shopping easier and save time.
